1. Basic tests

Theorem 1.1. *Case 1: Environment ends with simple text.* ♡

Theorem 1.2. *Case 2: Environment ends with display*

$$1 + 1 = 2. \qquad \heartsuit$$

Theorem 1.3. *Case 3: Environment ends with equationlike (numbered)*

$$1 + 1 = 2. \tag{1}$$ ♡

Theorem 1.4. *Case 4: Environment ends with*

- *a*
- *list.* ♡

2. Tests with multiple or nested environments

These test multiple or nested boxes inside of theoremlike environments.

Remark 2.1. Case 1: Environment ends with simple text. But first there is an equation

$$1 + 1 = 2 \tag{2}$$

and then the simple text. ◇

Remark 2.2. Case 2: Environment ends with display

(1) but first
(2) a list

$$1 + 1 = 2. \qquad \diamond$$

Remark 2.3. Case 3: Environment ends with align following

- a
- list

and

`display math`

`again`

$$x = 2 \tag{3}$$

$$y = 3. \quad \text{this is a very wide box that uses many words to fill the line} \tag{4}$$ ◇

Remark 2.4. Case 4: Environment ends with

- a
- list.
- that is nested
  (1) inside another
  (2) list ◇

Note 2.1. Case 4: Environment ends with

- a
- list
- with a display

$$a^2 + b^2 = c^2$$

  (1) and there is

    (2) a nested list
- but the outer list has
- the final item $\bullet$

Note 2.2.

- a
- list
    (1) and there is
    (2) a nested list
    (3) with a display
$$a^2 + b^2 = c^2 \qquad \bullet$$

Note 2.3.

- a
- list
    (1) and there is
    (2) a nested list
    (3) with align

$$a^2 + b^2 = c^2 \tag{5}$$
$$3^2 + 4^2 = 5^2 \tag{6}$$
$\bullet$

Note 2.4.

- a
- list
    (1) and there is
    (2) a nested list
    (3) with align

$$a^2 + b^2 = c^2 \tag{7}$$
$$3^2 + 4^2 = 5^2 \tag{8}$$

Then there is some text at the end $\bullet$

Note 2.5. First an align

$$a^2 + b^2 = c^2 \tag{9}$$
$$3^2 + 4^2 = 5^2 \tag{10}$$

Then some text,

(1) and then
(2) a list. $\bullet$

2.1. **some other messy tests.**

Theorem 2.1. *This ends with*

(1) *an*
(2) *enumerated*
(3) *list.* $\heartsuit$

Theorem 2.2. *This ends with*

(1) *an*
(2) *enumerated*

(3) *list*
(4) *ending in equation:*

$$a = b \qquad \heartsuit$$

Theorem 2.3. *This ends with*

(1) *an*
(2) *enumerated*
(3) *list.*
    (a) *abc*
(4) *blub* ♡

Theorem 2.4. *This ends with*

(1) *an*
(2) *enumerated*
(3) *list.*
    (a) *abc* ♡

2.2. **proof tests.**

Proof. regular text □

Proof. This ends in a display

$$3^2 + 4^2 = 5^2. \qquad \square$$

Proof. This ends in a display

$$3^2 + 4^2 = 5^2.$$

more text

$$a = b \tag{11}$$

$$c = d \qquad \square \tag{12}$$

Proof. This ends in a display

$$3^2 + 4^2 = 5^2.$$

more text

$$a = b$$

abc □

Proof. This ends with

(1) an
(2) enumerated
(3) list. □

Proof. This ends with

(1) an
(2) enumerated
(3) list
(4) ending in equation:

$$a = b \qquad \square$$

Proof. This ends with

(1) an

(2) enumerated
(3) list.
    (a) abc
(4) blub □

Proof. This ends with

(1) an
(2) enumerated
(3) list.
    (a) abc □

3. Tests of nested theoremlike environments

Not officially supported, but sort of works

Theorem 3.1. *regular text*

Lemma 3.1. *blub; lemma is not endmarked*

Lemma 3.2. *abc*

abc

(1) *abc*
    - *blub*
(2) *jhdgd* ♡

Text following theorem environment. *As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason.*

Theorem 3.2. *regular text*

Remark 3.1. blub; remark *is* endmarked ◇

♡

Text following theorem environment. *As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason.*

Theorem 3.3. *regular text*

Remark 3.2. blub; remark *is* endmarked ◇

Lemma 3.3. *abc; lemma is not endmarked*

abc

(1) *abc*
    - *blub*
(2) *jhdgd* ♡

Text following theorem environment. *As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason.*

4. Math environments

Subsections are intended to group environment types according to how they are handled internally; revise as necessary.

Many of the examples and descriptions are from the amsmath documentation: Section 3, Displayed equations.

4.1. equationlike.

These environments are full line width. They do not use horizontal alignment markers (&).

Test 4.1. `equation`: a single equation with an automatically generated number

$$\int_a^b f(x)\,dx \tag{13}$$ ◇

Test 4.2. `equation*`: unnumbered variant

$$\int_a^b f(x)\,dx$$ ◇

Test 4.3. `\[..\]`: The classic display math. Internally, amsmath redefines this as equation*

$$\int_a^b f(x)\,dx$$ ◇

Test 4.4. `multline`: a variation of equation, for long equations with linebreaks. The first line is aligned left, middle lines are centered, last line is aligned right. Equation number is on the first line; end mark is on the last line.

$$\begin{multline} a+b+c+d+e+f+a+b+c+d+e+f \\ +g+h+g+h+g+h+ \\ +i+j+k+l+m+n+i+j+k+l+m+n \tag{14} \end{multline}$$ ◇

Test 4.5. `multline*`: unnumbered variant

$$\begin{multline*} a+b+c+d+e+f+a+b+c+d+e+f \\ +g+h+g+h+g+h+ \\ +i+j+k+l+m+n+i+j+k+l+m+n \end{multline*}$$ ◇

Test 4.6. `multline`: a variation of equation, for long equations with linebreaks. The first line is aligned left, middle lines are centered, last line is aligned right. Equation number is on the first line; end mark is on the last line.

$$\begin{multline} abc \\ dfg \tag{15} \end{multline}$$ ◇

Test 4.7. `displaymath`: This is an older environment, not one of the amsmath ones; the amsmath documentation says it's functionally equivalent to equation*

$$\int_a^b f(x)\,dx$$ ◇

4.2. **alignlike.** These environments are full line width. They use horizontal alignment markers (&).

Test 4.8. `align`: for two or more equations; this test has just one alignment column

$$\int_a^b f(x)\,dx = 1 \tag{16}$$

$$\int_a^b g(x) + h(x)\,dx = a + b + c \tag{17}$$

◇

Test 4.9. `align`: another test, with multiple alignment columns

$$\begin{aligned} \int_a^b f(x)\,dx &= 1 & x &= y + z & p &= 0 \qquad (18)\\ \int_a^b g(x) + h(x)\,dx &= a + b + c & x' + y' &= z' & q &= 1 \qquad (19) \end{aligned}$$

◇

Test 4.10. `align*`: unnumbered variant; one column

$$\begin{aligned} \int_a^b f(x)\,dx &= 1 \\ \int_a^b g(x) + h(x)\,dx &= a + b + c \end{aligned}$$

◇

Test 4.11. `align*`: unnumbered variant; multiple columns

$$\begin{aligned} \int_a^b f(x)\,dx &= 1 & x &= y + z & p &= 0 \\ \int_a^b g(x) + h(x)\,dx &= a + b + c & x' + y' &= z' & q &= 1 \end{aligned}$$

◇

Test 4.12. `alignat`: allows the horizontal space between equations to be explicitly specified; takes an argument for number of "equation columns".

$$\begin{alignedat}{2} x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)} \qquad (20)\\ &= y' \circ y^* && \text{by (D)} \qquad (21)\\ &= y(0)y' && \text{by Axiom 1.} \qquad (22) \end{alignedat}$$

◇

Test 4.13. `alignat*`: unnumbered variant

$$\begin{alignedat}{2} x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)} \\ &= y' \circ y^* && \text{by (D)} \\ &= y(0)y' && \text{by Axiom 1.} \end{alignedat}$$

◇

Test 4.14. `gather`: a group of numbered equations with no horizontal alignment; each line is horizontally centered.

$$\int_a^b f(x)\,dx \tag{23}$$

$$\int_a^b f(x)g(x)h(x)\,dx \tag{24}$$

◇

Note: internally, `gather` and `gather*` are handled the same way as `align`, with the other alignlike environments, even though `gather` doesn't use alignment markers (&).

Test 4.15. `gather*`: unnumbered variant

$$\int_a^b f(x)\,dx$$
$$\int_a^b f(x)g(x)h(x)\,dx$$

◇

5. Lists

Test 5.1. `enumerate`: items are counted

(1) top
(2) middle
(3) bottom

◇

Test 5.2. `itemize`: items are not counted

- top
- middle
- bottom

◇

Test 5.3. `description`: items have labels

The first one:: goes here
Another:: and this is the item content
The final one:: these are often used for defintions

◇

Test 5.4. `list`: This is the basic environment that others are derived from.

first
middle
last

◇

6. Other blocks

These other special cases are each handled separately.

Test 6.1. `quote`:

> A quotation goes here. *As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason.*

◇

Test 6.2. `center`: center the contents

some centered text, or equation, or other content

◇

Note: the final line is off-centered, to make room for the endmark. It is usually unnoticeable, except in cases with visually similar lines, such as the following.

Test 6.3. `center`: center the contents

some centered text, or equation, or other content
some centered text, or equation, or other content
some centered text, or equation, or other content ◇

In cases like the above, some manual adjustment is necessary. For example, some negative `\hspace` for the width of the end mark.

Alternatively, an extra line break can move the endmark to a new line. This is shown in the next example.

Test 6.4. `center`: center the contents

some centered text, or equation, or other content
some centered text, or equation, or other content
some centered text, or equation, or other content

◇

6.1. **tabbing.** See source2e: File 41 ltab.dtx: Tabbing, Tabular, and Array Environments.

Explanation and examples from `https://latexref.xyz`.

Test 6.5. `tabbing`: manually set and use tab stops

row1-col1	row1-col2	this one has a longer heading	r1-c4
row2-col1	row2-col3	.	.
r3-c1	next field	a	b ◇

Test 6.6. `tabbing`: manually set and use tab stops

row1-col1	row1-col2	this one has a longer heading	r1-c4
row2-col1	row2-col3	.	.
r3-c1	next field	a	longer entry 012

◇

Above: the final longer entry pushes the end mark down.

6.2. **verbatim.** Internally, this is handled with the same code that handles lists (trivlist). Currently this only sort of works. The `\end{verbatim}` must be separated by a space, but not a newline, from the contents. This could have to do with the `hmode` v.s. `vmode` checks in the thmmark code.

Test 6.7. `verbatim`: Note that linebreaks are also treated verbatim.

```
Some verbatim symbols \ & $ etc.
More verbatim symbols \ & $ etc. More verbatim symbols \ & $ etc. More verbatim symbols \ & $
(The previous line intentionally runs off the page)
```
◇

Test 6.8. `verbatim`: (Another test) If the last line is *just a little* too long, it will push the endmark into the margin.

```
A123456789.B123456789.C123456789.D123456789.E123456789.F123456789
```
◇

Test 6.9. `verbatim`: (Another test) If the last line is *a little more* too long, it will push the endmark down.

```
A123456789.B123456789.C123456789.D123456789.E123456789.F123456789.
```
◇

7. Unsupported

These environments are currently unsupported. The marks may or may not be placed as expected, depending on the surrounding environment(s).

7.1. **Unsupported math environments.**

Test 7.1. `$$..$$`: This syntax is still used for display math. By default, it doesn't work well with `qedhere`; the end mark is not pushed to end of line.

$$\int_a^b f(x)\,dx$$

◇

Test 7.2. `eqnarray`: not recommended, for many reasons documented in Avoid `eqnarray`! (TUGboat 33:1, 2012)

$$\int_a^b f(x)\,dx \quad = 1. \tag{25}$$

◇

Test 7.3. `flalign`: full length alignment; stretches to fill line, leaving space for equation numbers and endmark (if present)
Currently unsupported. Manual placement of via `qedhere` works fine, but automatic placement with our alignlike code causes overlap of end mark with content.

$$\begin{aligned} x &= y & X &= Y \qquad (26)\\ x' &= y' & X' &= Y' \qquad (27)\\ x + x' &= y + y' & X + X' &= Y + Y' \qquad (28) \end{aligned}$$

◇

Test 7.4. `flalign*`: unnumbered variant; as with the numbered variant, manual placement treats endmark correctly, but automatic does not

$$\begin{aligned} x &= y & X &= Y\\ x' &= y' & X' &= Y'\\ x + x' &= y + y' & X + X' &= Y + Y' \end{aligned}$$

◇

7.2. **Math environments that go inside others.** The width of the following environments is just the width of the content. Each of these examples is inside `equation`. Each takes an optional argument for vertical positioning (baseline); demonstrated just with the first one.

Test 7.5. `gathered`: [c] (the default)

$$\begin{gathered} \int_a^b f(x)\,dx\\ \int_a^b f(x)\,dx\\ \int_a^b f(x)\,dx\\ a + b + c + d = e + f + g + h + i + j + k \end{gathered} \tag{29}$$

◇

Test 7.6. `gathered`: [t] (baseline top)

$$\begin{gathered}[t] \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ a+b+c+d=e+f+g+h+i+j+k \end{gathered} \tag{30}$$ ◇

Test 7.7. `gathered`: [b] (baseline bottom)

$$\begin{gathered}[b] \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ a+b+c+d=e+f+g+h+i+j+k \end{gathered} \tag{31}$$ ◇

Test 7.8. `aligned`:

$$\begin{aligned} \int_a^b f(x)\,dx &= 2 \\ \int_a^b f(x)\,dx &= x+y+z \\ \int_a^b f(x)\,dx &= 1 \\ a+b+c+d &= e+f+g+h+i+j+k \end{aligned} \tag{32}$$ ◇

Test 7.9. `alignedat`:

$$\begin{alignedat}{2} x &= y_1 - y_2 + y_3 - y_5 + y_8 - \dots &\quad& \text{by (C)} \\ &= y' \circ y^* && \text{by (D)} \\ &= y(0) y' && \text{by Axiom 1.} \end{alignedat} \tag{33}$$ ◇

Test 7.10. `cases`:

$$P_{r-j} = \begin{cases} 0 & \text{if } r-j \text{ is odd,} \\ r!\,(-1)^{(r-j)/2} & \text{if } r-j \text{ is even.} \end{cases}$$ ◇

Test 7.11. `dcases`: This is a variant that the amsmath docs mention, but is provided by the `mathtools` package. It makes the case lines displaystyle instead of (the default) textstyle. So, maybe we don't want to support this directly (especially not if we decide not to make `cases` endmarkable in the first place).
[*no example here*] ◇

Test 7.12. `split`: for a *single* equation split among multiple lines, like `multline`, but with some alignment points.

This `split` is inside `equation`. Note: the end mark is missing.

$$H_c = \frac{1}{2n} \sum_{l=0}^{n} (-1)^l (n-l)^{p-2} \sum_{l_1+\dots+l_p=l} \prod_{i=1}^{p} \binom{n_i}{l_i} \cdot [(n-l)-(n_i-l_i)]^{n_i-l_i} \cdot \Big[(n-l)^2 - \sum_{j=1}^{p} (n_i-l_i)^2\Big]. \tag{34}$$

Test 7.13. `split`: This `split` is inside `gather`, and the endmark is present.

$$H_c = \frac{1}{2n} \sum_{l=0}^{n} (-1)^l (n-l)^{p-2} \sum_{l_1+\dots+l_p=l} \prod_{i=1}^{p} \binom{n_i}{l_i} \cdot [(n-l)-(n_i-l_i)]^{n_i-l_i} \cdot \Big[(n-l)^2 - \sum_{j=1}^{p} (n_i-l_i)^2\Big]. \tag{35}$$ ◇

Note: The endmark is present in one `split` example, but not the other. This seems like a bug.

Test 7.14. `pmatrix`: other matrix environments are likewise not (directly) supported: `smallmatrix`, `bmatrix`, `Bmatrix`, `vmatrix`, and `Vmatrix`. Just let the surrounding environment handle the end mark.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$ ◇

Test 7.15. `array`: works inside displayed math; this one is in equation*. Just let the surrounding environment handle the end mark.

$$\chi(x) = \left| \begin{array}{ccc} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{array} \right|$$ ◇

Others. A few more examples that are often wrapped in other environments such as `center`.

tabular.

Test 7.16. `tabular`: a table; wrap in another environment to position endmark

Player name	*Career home runs*
Hank Aaron	755
Babe Ruth	714
Hank Aaron (repeat)	755
Babe Ruth (repeat)	714

◇

Test 7.17. `tabular`: a table in a `center`, with experimental command `\qedabove` to manually position the endmark

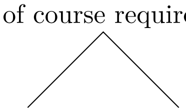
Player name	*Career home runs*
Hank Aaron	755
Babe Ruth	714
Hank Aaron (repeat)	755
Babe Ruth (repeat)	714

$\diamond$

Note: `\qedabove` can mess up spacing of the following lines, such as this one. *As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason.*

tikz environments. If these are wrapped in other environments like `equation`, then they won't need direct support

Test 7.18. `tikzpicture`: often used for diagrams, but of course requires tikz



$\diamond$

Test 7.19. `tikzcd`: another popular one, also requires extra packages
This is a `tikzcd` inside an `equation`.

$$\begin{tikzcd} A \arrow[r, "\phi"] \arrow[d] & B \arrow[d, "\psi"] \\ C \arrow[r, "\eta"] & D \end{tikzcd} \tag{36}$$

$\diamond$

8. Other Tests

Here are experimental tests of two things:

(1) ways to disable the thmmark for one box and
(2) ways to put the thmmark on a previous line.

These are buggy and incomplete.

Test 8.1. Use `\par` and negative `\vspace` to manually set endmark at bottom of environment. Note: This will cause overlap with content, if the previous line is too long, so it should only be placed manually.

$$\chi(x) = \begin{vmatrix} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{vmatrix}$$

◇

Paragraph after environment, to check spacing. *Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories.*

Test 8.2. (duplicate, with no extra commands, to check spacing)

$$\chi(x) = \begin{vmatrix} x-a & -b & -c \\ -d & x-e & -f \\ -g & -h & x-i \end{vmatrix} \qquad \diamond$$

Paragraph after environment, to check spacing. *Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories.*

Test 8.3. `\noqed` does not work here, at all

$$x = 2 \tag{37}$$

$$y = 3 \tag{38}$$

$$P_{r-j} = \begin{cases} 0 & \text{if } r-j \text{ is odd,} \\ r!\,(-1)^{(r-j)/2} & \text{if } r-j \text{ is even} \\ \text{other} & \text{case, for testing.} \end{cases} \tag{39}$$

a line of text ◇

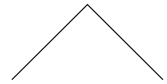
Test 8.4. `\noqed` does not work here

$$x = 2 \tag{40}$$

$$y = 3 \qquad \diamond \tag{41}$$

8.1. **tikzpicture.** Some tests of `tikzpicture` in equations. This is how I (NGJ) often make numbered commutative diagrams.

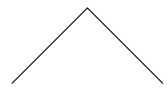
Test 8.5. `tikzpicture` in `equation`
baseline (0,0)



(42) $\diamond$

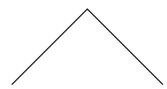
Test 8.6. another `tikzpicture` in `equation`
baseline (1,1)

(43) $\diamond$



Test 8.7. another `tikzpicture` in `equation`
baseline (1,1); use `\par` and negative `\vspace` to reposition endmark

(44)



$\diamond$

9. Tall lines in align and friends

These tests demonstrate end mark placement when an `align` or similar environment has a very tall line. By default, the end mark is placed on the baseline, aligned with an equation number, if present. Some of these tests experiment with different ways to manually adjust the vertical alignment.

Test 9.1. `align`: with `cases`

$$\text{words to push this content to the right edge} \qquad x = 2 \tag{45}$$

$$y = \begin{cases} \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \end{cases} \tag{46}$$ ◇

Spacing test. *As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions.*

Test 9.2. Similar to above, but manually reposition endmark with
`\par\vspace{-1.6\baselineskip}~`
after `\end{align}`

$$\text{words to push this content to the right edge} \qquad x = 2 \tag{47}$$

$$y = \begin{cases} \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \\ \int_a^b f(x)\,dx \end{cases} \tag{48}$$ ◇

What if there is another line of text? Is the endmark placed above?

Spacing test. *As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions.*

Notes for the previous example.

(1) Ending with `\qedabove` also works.
(2) Ending with `\qedhere` doesn't work; the endmark is on the baseline.
(3) Ending with `\qedhere` (as in `\qedprev`) followed by `\hspace` *does* work; the endmark appears after the fourth `cases` row.
    It doesn't matter how much `hspace`, as long as it's a nonzero amount. This is probably something to do with `vmode` v.s. `hmode`, and if so then there is probably a better way to handle it. That's beyond what I currently understand though.

(4) Spacing after the environment doesn't seem to be affected by the negative `vspace`, but lines *inside* the environment are affected.
(5) Maybe the whole negative `vspace` idea is just not great; is there a better alternative?